

EXPERIMENT-19:

Implementation of Naive Bayes classification for Bank Loan prediction.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Upload Dataset
print("Upload train.csv")
uploaded = files.upload()

# Read Uploaded File
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)

# Display Dataset
print("\nFirst 5 Rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nMissing Values:")
print(df.isnull().sum())

# Remove Loan_ID column if present
if 'Loan_ID' in df.columns:
    df = df.drop('Loan_ID', axis=1)

# Fill Missing Values
for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = df[col].fillna(df[col].mode()[0])
    else:
        df[col] = df[col].fillna(df[col].median())
```

```

# Convert Categorical Columns into Numeric
encoder = LabelEncoder()
for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = encoder.fit_transform(df[col])
# Features and Target
X = df.drop('Loan_Status', axis=1)
y = df['Loan_Status']
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
# Train Naive Bayes Model
model = GaussianNB()
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\n=====")
print("Model Accuracy")
print("=====")
print(f'{accuracy*100:.2f}%')
# Classification Report
print("\n=====")
print("Classification Report")
print("=====")

```

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print(classification_report(y_test, y_pred))

# Confusion Matrix

print("\n=====")

print("Confusion Matrix")

print("=====")

print(confusion_matrix(y_test, y_pred))

# Predict First Applicant

sample = X.iloc[[0]]

prediction = model.predict(sample)[0]

print("\n=====")

print("Prediction")

print("=====")

if prediction == 1:

    print("Loan Status : Approved")

else:

    print("Loan Status : Rejected")

```

Output:

```

... Upload train.csv
  Choose Files train_u6luju...tuZ9i[1].csv
train_u6lujuX_CVtuZ9i[1].csv(text/csv) - 38013 bytes, last modified: 8/7/2026 - 100% done
Saving train_u6lujuX_CVtuZ9i[1].csv to train_u6lujuX_CVtuZ9i[1].csv

First 5 Rows:
   Loan_ID Gender Married Dependents Education Self_Employed \
0 LP001002 Male      No           0 Graduate             No
1 LP001003 Male      Yes           1 Graduate             No
2 LP001005 Male      Yes           0 Graduate             Yes
3 LP001006 Male      Yes           0 Not Graduate         No
4 LP001008 Male      No            0 Graduate             No

   ApplicantIncome CoapplicantIncome LoanAmount Loan_Amount_Term \
0           5849           0.0           NaN           360.0
1           4583          1508.0          128.0           360.0
2           3000           0.0           66.0           360.0
3           2583          2358.0          120.0           360.0
4           6000           0.0          141.0           360.0

   Credit_History Property_Area Loan_Status
0             1.0         Urban           Y
1             1.0         Rural           N
2             1.0         Urban           Y
3             1.0         Urban           Y
4             1.0         Urban           Y

Dataset Shape:
(614, 13)

```

```
... Missing Values:
Loan_ID          0
Gender           13
Married          3
Dependents       15
Education        0
Self_Employed   32
ApplicantIncome  0
CoapplicantIncome 0
LoanAmount       22
Loan_Amount_Term 14
Credit_History  50
Property_Area    0
Loan_Status      0
dtype: int64
```

```
=====
Model Accuracy
=====
84.55%
```

```
=====
Classification Report
=====
```

	precision	recall	f1-score	support
0	0.88	0.58	0.70	38
1	0.84	0.96	0.90	85
accuracy			0.85	123
macro avg	0.86	0.77	0.80	123
weighted avg	0.85	0.85	0.84	123

```
=====
Confusion Matrix
=====
```

[[22 16]
[3 82]]

```
=====
Prediction
=====
Loan Status : Approved
```